WEAK UNRAMIFIEDNESS IS PRESERVED BY A TOTALLY RAMIFIED SEPARABLE BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

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Editorial note. The counted statement span contains the author's displayed Eisenstein polynomial in the proof span and references the construction remark for the meaning of A_1 and B_1; both are delivered verbatim. The single reference inside the counted proof span is recorded as a non-load-bearing navigation pointer.

Allowed standard prerequisites (editorial).

- Totally ramified separable extensions and the Eisenstein minimal polynomial of a uniformizer. used at proof lines 34501-34509. Supplies the presentation B_1 = B[t]/(P(t)) with the uniformizer.
- Weak unramifiedness implying that the uniformizer is preserved. used at proof lines 34510-34511. Is what makes the displayed ring a discrete valuation ring over B.

Source reference. Original remark 33882: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33882–33882. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $A \to B$ be an extension of discrete valuation rings with fraction fields $K \subset L$. Assume that $A \to B$ is weakly unramified. Then for any finite separable extension K_1/K totally ramified with respect to A we have that $L_1 = L \otimes_K K_1$ is a field, A_1 and $B_1 = B \otimes_A A_1$ are discrete valuation rings, and the extension $A_1 \subset B_1$ (see Remark 33882) is weakly unramified.*

Proof. Let $\pi \in A$ and $\pi_1 \in A_1$ be uniformizers. As K_1/K is totally ramified with respect to A we have $\pi_1^e = u_1\pi$ for some unit u_1 in A_1. Hence A_1 is generated by π_1 over A and the minimal polynomial $P(t)$ of π_1 over K has the form

$$P(t) = t^e + a_{e-1}t^{e-1} + \ldots + a_0$$

with $a_i \in (\pi)$ and $a_0 = u\pi$ for some unit u of A. Note that $e = [K_1 : K]$ as well. Since $A \to B$ is weakly unramified we see that π is a uniformizer of B and hence

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$B_1 = B[t]/(P(t))$ is a discrete valuation ring with uniformizer the class of t. Thus the lemma is clear. $\square$